

GeoGuessrNet: Country-Level Image Geolocation with CNNs and Vision Transformers

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Abstract

Determining the geographic location of an image using only visual input is a complex yet compelling challenge with wide-ranging applications in journalism, disaster response, and content verification. In this work, we introduce a country-level image geolocation framework that benchmarks three distinct approaches: human inference, a vision-language large model (GPT-4o), and two deep-learning models fine-tuned for this task—a convolutional neural network (ResNet-50) and a Vision Transformer (ViT-B/16). We train and evaluate each method on a 50,000-image dataset derived from the GeoGuessr game, which spans 124 countries and presents substantial visual and geographic diversity.

Our results reveal that while GPT-4o—without any task-specific tuning—achieves the highest accuracy (71.5% top-1), fully fine-tuned ResNet-50 (41.3% top-1), and fully fine-tuned ViT-B/16 (67.5% top-1) models substantially outperform human participants (13.2% top-1). We also show that ViT models, which capture global contextual features more effectively than CNNs, offer even greater performance gains. Overall, we demonstrate that transformer-based architectures can rival multimodal giants and far exceed human intuition in country-level image geolocation.

1. Introduction

Determining the geographic location of an image is a challenging yet fundamentally human task. Humans naturally leverage a variety of visual cues—such as climate characteristics, architectural styles, languages on signs, types of vegetation, and even subtle road markings—to pinpoint where an image was taken. Automating this process using deep learning has significant implications, enabling applications that range from verifying the authenticity of social media content to facilitating rapid response efforts during humanitarian emergencies.

Our motivation for this project stems from the popular online game *GeoGuessr*, where players are challenged to identify the location of street-view images from around the globe. The inherent difficulty of this game underscores the complexity of geographic classification; even experienced players frequently confuse visually similar regions—for example, mistaking southern Brazil for northern Portugal due to similarities in landscape and architecture. Inspired by this challenge, we aim to build an effective deep-learning model capable of reliably identifying the country where an image was captured, thereby minimizing common human errors and enhancing the interpretability of the decision-making process.

Formally, we frame our problem as follows: given an RGB image $I \in \mathbb{R}^{H \times W \times 3}$, our task is to predict the country label $y \in 1, \dots, K$ where the image was captured. Specifically, the input to our algorithm is a single RGB image. We employ different techniques, primarily Vision Transformers (ViT), to output a predicted country label. To comprehensively evaluate our model’s effectiveness, we also implement and benchmark performance against a CNN ResNet, an LLM baseline using GPT, and human performance. These baselines establish performance thresholds, and contextualize the improvement and insight our ViT model achieves in geographic image classification.

By targeting the country-level granularity, we balance computational efficiency and actionable insight, making our approach practical for downstream applications such as regional trend analysis, location-based content moderation, and real-time geographic inference tasks. Ultimately, our goal is to demonstrate that models like ViT can significantly surpass both human intuition and conventional model baselines, thus advancing the state-of-the-art in automated geographic location identification.

2. Related Work

Early attempts at global image geolocation largely relied on traditional computer vision methods involving hand-

crafted features and nearest-neighbor search. Notably, Im2GPS [6] used color histograms, texture descriptors, and other image-level statistics to retrieve similar images from a database, predicting locations by finding the closest matches. While groundbreaking at the time, this method had limitations in scalability and generalizability due to the reliance on manually designed features and nearest-neighbor retrieval.

The rise of convolutional neural networks (CNNs) significantly advanced the accuracy and scalability of geolocation methods. A landmark contribution was the PlaNet system [13], which divided Earth into approximately 26,000 discrete geocells, treating geolocation as a classification problem. PlaNet showcased the effectiveness of CNNs by outperforming traditional approaches significantly, reaching city-level accuracy through massive-scale training. Despite its strengths, PlaNet’s accuracy was strongly dependent on dataset coverage and remained computationally expensive due to the granularity of prediction.

Recent studies shifted towards predicting at coarser granularities such as the country level, offering a good compromise between accuracy and computational complexity. Almayreh *et al.* [1] proposed *VIPPGeo*, demonstrating that directly predicting country labels using a ResNet architecture outperforms approaches that regress GPS coordinates and subsequently map them to country polygons. Further building on country-level classification, Haas *et al.* introduced the PIGEON model[5], which implements semantic geocells, grouping visually similar areas and significantly improving model interpretability. These works demonstrated that country-level classification effectively balances prediction accuracy and practical application scenarios.

Transformers, particularly Vision Transformers (ViTs), have recently emerged as powerful alternatives to CNNs due to their ability to capture global context effectively. Pramanick *et al.* [10] employed ViTs to geolocate images taken in diverse real-world conditions, showcasing transformers’ superior ability to integrate context across distant image regions. Vivanco *et al.*’s GeoCLIP [12] leveraged multimodal transformer models to align geographic embeddings of locations with visual features, enhancing global geolocation accuracy. Furthermore, StreetCLIP [4] demonstrated that transformers pretrained on language-image pairs could perform impressive zero-shot localization, emphasizing the flexibility and generalizability of transformer-based architectures.

Multimodal approaches, integrating additional context beyond purely visual data, have further refined image geolocation techniques. Luo *et al.* introduced the G³ model[8], which utilized textual guidebook data to enrich image representations, demonstrating substantial accuracy gains through cross-modal grounding. Similarly, GeoCLIP [12] introduced contrastive learning methods to align images and

geographic information, resulting in robust generalization to unseen locations. While these multimodal strategies significantly advance geolocation accuracy, their reliance on external text data can limit applicability when such meta-data is unavailable.

These collective insights from previous work motivate our comparative study of CNN-based ResNet architectures and ViT backbones for the task of country-level image classification. Moreover, they underscore the potential value of future multimodal extensions to further enhance geolocation accuracy and interpretability.

3. Data

We use the publicly available GeoGuessr images dataset from Kaggle [11], consisting of 49,997 street-level RGB screenshots captured directly from the GeoGuessr game. Each image has a resolution of 1536×662 pixels and is labeled with the country in which it was taken. The dataset spans a total of 124 unique countries, though distribution across these countries is highly uneven, ranging from just 1 image to a maximum of 12,014 images per country. On average, each country has approximately 403 images, but the median is significantly lower at 81 images per country, highlighting the imbalance in geographical representation.

Given the unbalanced nature of this dataset, we perform stratified splits to maintain a fair representation of countries across training, validation, and test subsets. Specifically, we apply an 80/10/10 split for train, validation, and test sets, respectively. For countries with only 1 image, we omit their instances from the validation and test sets to avoid biasing model evaluation due to inadequate sample representation.

The dataset itself consists of street-level scenes captured by players during actual GeoGuessr gameplay. Consequently, each image contains several sources of noise characteristic of the GeoGuessr interface, including navigation arrows, a scorecard positioned in the top-right corner, a blurred map in the bottom-right corner, and a compass indicator in the bottom-left corner (see Fig. 1). This presents an additional challenge for our models, as they must generalize beyond interface-related visual clutter to identify relevant geographic features.

The scenes themselves vary tremendously in content, ranging from easily identifiable city streets, signage featuring specific languages, and characteristic vegetation or architectural styles, to images that provide minimal visual context, such as generic skies, rural landscapes, or expansive fields with sparse identifying details. Such variability closely mirrors the complexity humans face when playing GeoGuessr, making it an authentic representation of real-world geolocation challenges.

Minimal preprocessing steps were applied to the raw data beyond the described splitting methodology. We did not perform cropping or explicit removal of interface ele-

ments, preserving the realistic complexity inherent in each image. Similarly, no explicit data augmentation or normalization was performed prior to input to the models because techniques such as image rotation or flipping would distort critical geographic cues. In geolocation tasks, elements like the direction of road signs, the position of the sun, and text alignment provide important spatial context. Thus, data augmentation could compromise these directional cues and mislead the model.

Fig.1, Fig.2, Fig.3, and Fig.4 present representative examples from four geographically and visually distinct countries: the United States, Brazil, South Africa, and Antarctica. These examples highlight the wide range of environments captured in the dataset—from urban infrastructure and vegetation to rural landscapes and extreme climates—and illustrate the variety of visual cues available for country-level classification.



Figure 1: Example from the United States.



Figure 2: Example from Brazil.



Figure 3: Example from South Africa.



Figure 4: Example from Antarctica.

4. Methods

4.1. Human Baseline

To establish a meaningful point of comparison grounded in real-world performance, we collected a human baseline using five participants familiar with the GeoGuessr game. This included the three members of our research team and two additional volunteers. Each participant was shown the same set of 100 randomly sampled images from the dataset and asked to guess the country of origin for each image. These images were representative of the dataset’s diversity in geography, scene type, and difficulty, and were not included in any model’s training set. Participants were instructed to provide two responses for each image: (1) their single best guess for the country, and (2) a ranked list of their top five most likely countries. The goal of including a human baseline is to contextualize model performance against the way actual players engage with the geolocation task—leveraging high-level cues such as language, road signs, terrain, and architectural style. However, unlike a neural network trained across tens of thousands of labeled examples, human inference is highly limited in its consistency and regional knowledge, particularly in visually ambiguous or underrepresented regions.

4.2. LLM Baseline

To establish a performance upper bound, we evaluate OpenAI’s GPT-4o—one of the most advanced vision-language models currently available—on our test set. This model integrates a state-of-the-art vision encoder with a powerful large language model, enabling it to reason over complex visual scenes using knowledge acquired from large-scale Internet image-text data. For each image in the test set, we query the GPT-4o Vision endpoint with a prompt asking for the most likely country of origin, along with a ranked list of the top five predicted countries. The prompt is designed to elicit a structured JSON response for ease of downstream parsing and evaluation. Because GPT-4o has been trained on an enormous corpus of general-purpose multimodal data and benefits from extensive compute resources, its performance represents an approximate upper bound for this task. Crucially, the model receives no task-specific fine-tuning or domain adaptation for the GeoGuessr context, yet its ability to make high-quality predictions showcases the latent geographic reasoning capabilities learned from general visual-textual pretraining. This baseline allows us to situate the performance of our finetuned models within the broader landscape of frontier AI systems.

4.3. Variation 1: Finetuned ResNet-50

Our first learned model baseline uses a convolutional neural network: specifically, a finetuned version of the ResNet-50 v1.5 architecture pretrained on Im-

geNet, available via Hugging Face under the identifier `microsoft/resnet-50` [9]. ResNet-50, introduced by He *et al.* [7], is a 50-layer deep residual network that enables training of very deep architectures through the use of identity-based skip connections. These residual connections allow gradients to flow more easily through the network during backpropagation, alleviating the vanishing gradient problem common in deep convolutional models. The v1.5 variant, adopted by NVIDIA, modifies the bottleneck blocks to apply the stride-2 operation in the 3×3 convolution instead of the first 1×1 , offering a small accuracy gain at the cost of a slight performance drop.

We initialize the model with pretrained ImageNet weights and replace the original 1,000-class classification head with a new linear classifier for 104 countries (classes), corresponding to the subset of countries with at least 2 images. This newly added head is randomly initialized. Importantly, we finetune the entire network, not just the classifier head. All convolutional layers remain trainable, enabling the model to adapt low-level feature extraction to the specific visual distributions of GeoGuessr imagery.

Let $I \in \mathbb{R}^{H \times W \times 3}$ be an input image, and $f_\theta(I) \in \mathbb{R}^{104}$ be the output logits, where θ denotes the model parameters. We train the model to minimize cross-entropy loss:

$$\mathcal{L}_{\text{CE}} = - \sum_{k=1}^{104} \mathbb{I}[y = k] \cdot \log \left(\frac{\exp(f_\theta(I)_k)}{\sum_{j=1}^{104} \exp(f_\theta(I)_j)} \right)$$

where y is the ground truth country label.

We used Hugging Face’s `Trainer` API to manage the training pipeline. Each image is passed through `AutoImageProcessor` for resizing, normalization, and tensor conversion. The dataset is split into the 80/10/10 breakdown for training, validation, and test sets.

Training is conducted for 5 epochs using the AdamW optimizer with a batch size of 32 and mixed-precision enabled (FP16) when supported. While our search was not exhaustive, we did perform limited hyperparameter tuning on the learning rate and the number of training epochs. We found that a learning rate of 5×10^{-5} and training for 5 full epochs yielded stable convergence and strong validation performance without overfitting.

4.4. Variation 2: Finetuned ViT

Our second learned model builds on a pure attention-based architecture: the Vision Transformer (ViT). Specifically, we finetune the `google/vit-base-patch16-224` model provided by Hugging Face [3], originally introduced by Dosovitskiy *et al.* [2]. ViT was the first architecture to show that transformers, previously dominant in NLP, can outperform convolutional neural networks on large-scale

image classification tasks when pretrained on sufficient data.

The core idea behind ViT is to treat an image as a sequence of non-overlapping patches, flattening and linearly projecting each 16×16 RGB patch into a token embedding. For an input image $I \in \mathbb{R}^{3 \times 224 \times 224}$, this results in $N = 196$ tokens (plus one special classification token). These tokens are passed through a stack of transformer encoder layers, each consisting of multi-head self-attention and feed-forward sublayers, producing contextualized representations. The output corresponding to the class token is then used for classification via a linear head.

We adapt this model to our 104-way country classification task by replacing the original ImageNet classification head with a new randomly initialized linear layer. This final layer maps the $D = 768$ -dimensional output of the [CLS] token to a 104-dimensional logits vector, which we train to minimize the cross-entropy loss.

We initialize the model with weights pretrained on ImageNet-21k, using Hugging Face’s `ViTForImageClassification` API. The data preprocessing pipeline mirrors that of our ResNet experiment: images are normalized and resized using the corresponding `AutoImageProcessor`. We perform the same 80/10/10 stratified split and use the same transformation function to convert images into PyTorch tensors with associated labels.

Unlike ResNet, the ViT architecture lacks built-in inductive biases like locality and translation equivariance, which can make training from scratch difficult. However, this flexibility becomes an asset during finetuning, especially for complex global patterns such as vegetation zones, road textures, and signage fonts that span long spatial ranges. We hypothesize that ViT’s global receptive field is particularly well suited for GeoGuessr-style image localization, where distant cues (e.g., sky color or road lines) may be more informative than local textures.

However, it is important to note that the dataset had a significant class imbalance. For example, countries like the United States had thousands of images, but countries like Aland had only a handful of images. To address this, we experimented with Focal Loss, a modification of the standard cross-entropy loss designed to reduce the relative loss for well-classified examples and focus more on hard, misclassified examples. Focal loss modifies the standard cross-entropy loss by introducing a modulating factor, where p_{y_i} is the predicted probability for the true class and γ is a focusing parameter.

$$\mathcal{L}_{\text{focal}} = -(1 - p_{y_i})^\gamma \log(p_{y_i})$$

Despite our hypothesis, we found that standard cross-entropy loss led to better overall accuracy.

We trained ViT for 10 epochs using the AdamW optimizer with a learning rate of 2×10^{-5} and batch size of 32. We enabled mixed-precision (FP16) to accelerate training and save memory. Throughout development, we exhaustively tuned hyperparameters using short training runs (1–2 epochs) on the training and validation splits. We experimented with learning rates in the range $[1 \times 10^{-5}, 1 \times 10^{-3}]$, and weight decay values in $[0.01, 0.1]$, ultimately selecting the configuration that led to the highest top-1 accuracy on the validation set after 2 epochs. We also tested both full-model finetuning and classifier-head-only training (via the `--freeze` flag) and found that full finetuning yielded better performance.

All layers of the transformer are updated during training, allowing ViT to adapt its patch-level attention weights and internal representations to the unique distribution of GeoGuessr images. This end-to-end optimization process was essential for handling dataset-specific artifacts such as navigation UI elements and low-resolution rural scenes, which differ from the clean object-centric images seen in ImageNet.

5. Experiments and Results

5.1. Evaluation Metrics

To evaluate model performance on the country classification task, we adopt two standard metrics: top-1 accuracy and top-5 accuracy. Top-1 accuracy measures the fraction of test images for which the model’s most confident prediction matches the ground truth label. Top-5 accuracy measures the proportion of test samples where the correct label appears anywhere among the model’s five highest-probability predictions. This metric is useful for capturing partially correct predictions. Formally, given N samples, model predictions $\hat{y}_i$, and ground truth labels y_i , the top-1, and top-5 accuracies are defined as:

$$\text{Top-1 Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\hat{y}_i = y_i]$$

$$\text{Top-5 Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[y_i \in \hat{Y}_i^{(5)}]$$

These metrics allow us to evaluate both exact match performance and broader semantic understanding across our baselines and finetuned models.

5.2. Results

5.2.1 Quantitative Results

Table 1 summarizes results for our human baseline, vision–language LLM (GPT-4o), fully finetuned ResNet-50, and fully finetuned ViT-B/16.

Model	Top-1	Top-5
Human Baseline (n=5)	0.132	0.316
GPT-4o Vision	0.715	0.865
ResNet-50 (5 epochs)	0.413	0.592
ViT-B/16 (10 epochs)	0.675	0.883

Table 1: Test set performance across models.

The human baseline—collected from five participants with familiarity in GeoGuessr—yields 13.2% top-1 and 31.6% top-5 accuracy across a shared set of 100 images. These values serve as a sanity check for task difficulty and illustrate the limits of unaided human visual reasoning in diverse, global settings.

Our strongest non-learned baseline is GPT-4o, which demonstrates remarkable zero-shot generalization: 71.45% top-1 and 86.45% top-5 accuracy on the test set. This performance, achieved without any task-specific finetuning, highlights the strength of large vision–language models trained on broad web-scale image–text data.

By contrast, our fully finetuned ResNet-50 model reaches 41.3% top-1 and 59.2% top-5 accuracy after 5 epochs of training. Although the CNN trails the GPT-4o reference by a wide margin, its performance is already three times higher than the human baseline and was achieved with relatively lightweight hyperparameter tuning. This suggests that there is considerable headroom available through longer schedules, optimized learning rates, and stronger augmentation regimes.

Our best-performing learned model is the Vision Transformer (ViT-B/16), which achieves 67.5% top-1 and 88.3% top-5 accuracy after 10 epochs. This model significantly outperforms the finetuned ResNet-50, underscoring the benefits of global self-attention. Unlike CNNs, which rely on local receptive fields, ViT excels at modeling long-range spatial dependencies—an advantage in geographic reasoning tasks that require interpreting distributed environmental cues. These results validate the effectiveness of transformer architectures for structured, spatially grounded prediction problems.

Overall, these results demonstrate a clear progression in geographic reasoning ability across humans, CNNs, and transformer-based architectures. The ViT-B/16 model closes much of the performance gap with GPT-4o, reaching comparable top-1 and better top-5 accuracy despite being trained on a single dataset with far less compute and no multimodal pretraining. This validates both our training pipeline and the promise of transformers as a scalable, domain-adaptable solution for visual geolocation. It also suggests that further tuning, data augmentation, or lightweight multimodal integration could bring task-specific ViTs to parity with web-scale vision–language models.

5.2.2 Qualitative Results

Beyond aggregate metrics, qualitative analysis reveals where different agents—human and model—succeed or struggle. GPT-4o performed best in countries with high digital visibility, such as the United States, India, Japan, and Western Europe. These countries tend to be overrepresented in large-scale internet image–text corpora, allowing vision–language models to implicitly learn geographic priors. This pattern is clearly illustrated in Figure 5, where top-1 accuracy is highest in the Global North and urbanized nations. In contrast, performance is noticeably lower in underrepresented or geographically ambiguous regions such as Central Africa.

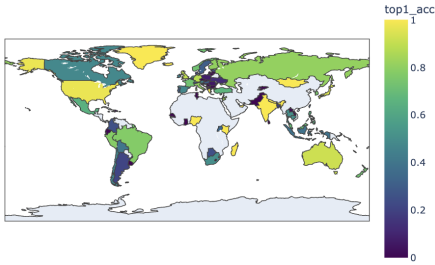


Figure 5: GPT-4o top-1 accuracy by country.

Human participants exhibited different strengths. While overall performance was significantly lower than the models, accuracy improved substantially when images contained visible text. Street signs, advertisements, and store names provided reliable cues for narrowing down the country. Additionally, all five participants correctly identified certain high-context images from the United States, suggesting that humans are especially effective at leveraging familiar visual landmarks and cultural knowledge. Figure 6 shows one such image that each human identified correctly. NYPD barricades, USPS logos, and American-style architecture made the country easily identifiable.



Figure 6: U.S. image that each human identified correctly.

To better understand where our ResNet-50 model succeeds and struggles, we analyzed its per-country top-1 and top-5 accuracy. The model performs best in countries with distinctive visual features or heavy representation in the training set, such as the United States, Japan, Australia, and the United Kingdom. Conversely, the model struggles

in countries with visual overlap—such as Balkan or Central American nations—or those with limited representation. This pattern reflects a core limitation of convolutional architectures: a reliance on local texture and object features without strong global context integration. The overall noisiness of the accuracy distribution across countries highlights that while ResNet captures some regional cues, it lacks the holistic scene reasoning necessary for more nuanced geographic disambiguation.

The ViT-B/16 model displays performance patterns that, while still correlated with data availability, reveal a more nuanced global generalization than either the human baseline or ResNet. As shown in Figure 7, ViT achieves high top-1 accuracy in countries with both high digital visibility and distinct visual environments—such as the United States (0.89), Japan (0.89), Germany (0.88), and India (0.83). These countries tend to have strong architectural identities, readable signage, or road features that ViT learns to associate with the correct label. The model also performs surprisingly well in countries with more limited internet presence, such as Botswana and Tunisia (both achieving 0.78+ top-1 accuracy), indicating that global attention mechanisms help ViT extract subtle environmental cues like vegetation, lighting, or road markings that are less reliant on cultural or urban infrastructure.

However, ViT still struggles in countries that are underrepresented in the dataset or that share close visual resemblance with others. The heatmap in Figure 8 reveals that accuracy drops sharply in the long tail of countries with few samples—e.g., Lithuania (0.14 top-1), Laos (0.17), and Ukraine (0.20). In some cases, such as Portugal and Colombia, top-5 accuracy remains relatively high while top-1 performance lags, suggesting the model is often directionally correct but cannot fully disambiguate among regional neighbors. Predictably, the model fails entirely in countries with extremely sparse or zero representation during training, such as Madagascar, Slovenia, and Uruguay. These limitations underscore the importance of data coverage and suggest that incorporating regional priors or training-time class reweighting could improve performance in low-resource settings.

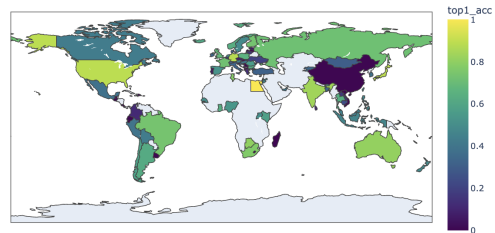


Figure 7: ViT-B/16 top-1 accuracy by country.

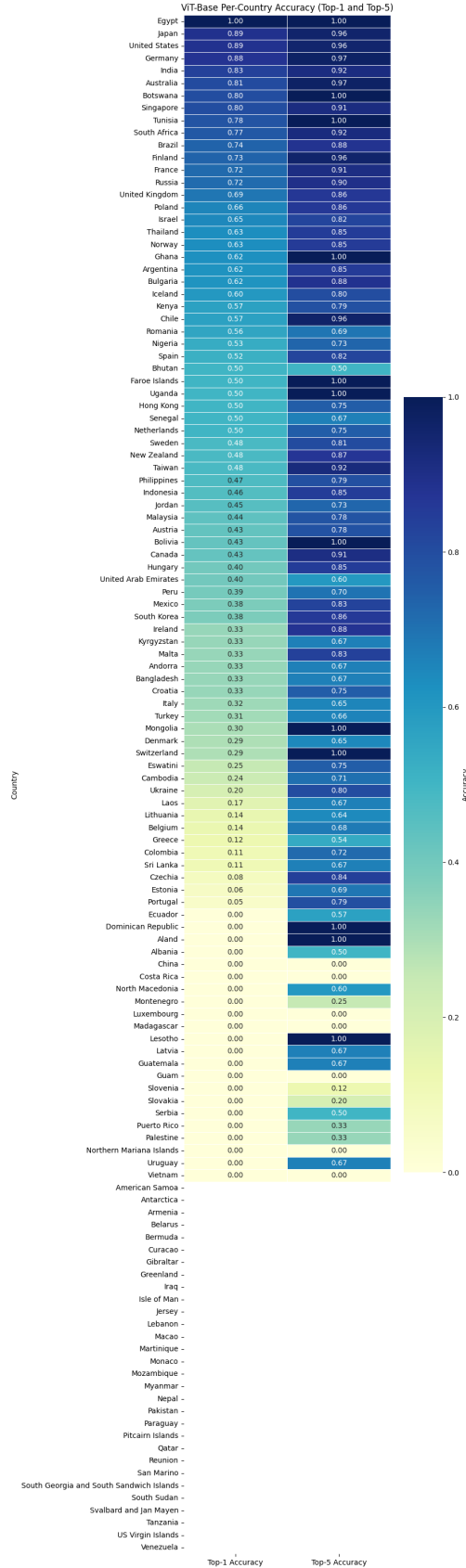


Figure 8: ViT-B/16 per-country top-1 and top-5 accuracy heatmap.

5.2.3 Limitations

A key limitation of this study lies in the dataset itself: countries with limited Google Street View coverage are under-represented or entirely absent, skewing the model’s exposure toward high-income or highly digitized regions. Additionally, our formulation restricts prediction to the country level, which obscures fine-grained errors and proves especially limiting for geographically large countries (e.g., Russia, Brazil) or visually similar neighboring regions (e.g., Belgium and the Netherlands). Finally, our human baseline is based on a small, homogeneous group ($n=5$), all of whom are American university students, which limits the generalizability of those results.

6. Conclusion

Our study illustrates the rapid progress that modern vision models have made on the long-standing challenge of image geolocation. Moreover, our experiments reveal a nuanced performance trade-off between frontier vision-language models and task-specific transformers. GPT-4o achieved the highest top-1 accuracy of 71.5 and barely beat our ViT-B/16 model. However, our ViT surpassed GPT-4o on top-5 accuracy (88.3% vs. 86.5%), which means that our model more reliably included the correct country somewhere in its shortlist of five guesses. An explanation for this is that GPT-4o is pretrained on vast web-scale image-text pairs and may have already seen many near-duplicate street-view scenes circulating online. This boosts its most-confident prediction but can leave it over-confident when the scene is genuinely novel. However, our ViT model had to learn solely from the GeoGuessr dataset, but it appears to better capture broader geographical possibilities even when its top guess is wrong.

This distinction matters for real-world deployment. In many geolocation tasks, such as fact-checking, disaster response, etc., an accurate top-5 candidate set can be more valuable than a single high-confidence guess. Our results show that a carefully finetuned ViT, trained on significantly less data and compute than GPT-4o, can already outperform the frontier model on this more practical metric. Closing the remaining top-1 gap will likely require richer, more balanced imagery and light multimodal cues, but the transformer’s ability to integrate long-range spatial context positions it as a competitive and more efficient alternative to massive vision-language systems.

6.1. Future Works

If given more time and compute resources, we would extend this project in several directions. First, we would conduct longer training schedules with stronger data augmentation and advanced learning rate schedulers (e.g., cosine annealing or one-cycle) to improve generalization in both

CNN and transformer models. Second, we would explore multimodal finetuning by adding synthetic captions or map metadata to better mimic GPT-4o’s information richness. Third, we are interested in developing hybrid architectures that combine ViT’s global attention with ResNet-style local feature extraction. Lastly, we would explore region-aware loss functions or curriculum learning strategies that dynamically emphasize underperforming regions—both to reduce model bias and improve coverage in visually ambiguous or low-resource countries.

7. Appendix

8. Contributions & Acknowledgments

8.1. Team Contributions

- Esi: Researched majority of Related Works, built most of ResNet model.
- Sungbin: Gathered and pre-processed data, performed most of the hyperparameter tuning and loss function experiments.
- Vikram: Wrote GPT baseline script, built most of ViT model, and performed data analysis/ made plots.
- Everyone contributed equally on paper.

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